

Answer Key: Quiz 2 - Introduction to Statistical Signal Processing

Q1 [8 Marks]

Given:

- $z[n] = x[n] + y[n]$
- $x[n]$ and $y[n]$ are zero-mean random processes ($E[x[n]] = E[y[n]] = 0$)

Part a) [6 Marks]

Given: The cross-correlation between x and y is $r_{xy}[n_1, n_2] = \sigma^2 \delta[n_1 - n_2]$.

Expansion and Simplification: Replacing $z[n]$ with $x[n] + y[n]$ and expanding the product:

$$\begin{aligned} r_z[n_1, n_2] &= E[z[n_1]z[n_2]] \\ &= E[(x[n_1] + y[n_1])(x[n_2] + y[n_2])] \\ &= E[x[n_1]x[n_2] + x[n_1]y[n_2] + y[n_1]x[n_2] + y[n_1]y[n_2]] \\ &= E[x[n_1]x[n_2]] + E[x[n_1]y[n_2]] + E[y[n_1]x[n_2]] + E[y[n_1]y[n_2]] \end{aligned}$$

In terms of correlation functions:

$$r_z[n_1, n_2] = r_x[n_1, n_2] + r_{xy}[n_1, n_2] + r_{yx}[n_1, n_2] + r_y[n_1, n_2]$$

Final Expression: Using $r_{yx}[n_1, n_2] = r_{xy}[n_2, n_1]$ and the symmetry of the delta function:

$$\begin{aligned} r_z[n_1, n_2] &= r_x[n_1, n_2] + r_y[n_1, n_2] + \sigma^2 \delta[n_1 - n_2] + \sigma^2 \delta[n_2 - n_1] \\ r_z[n_1, n_2] &= r_x[n_1, n_2] + r_y[n_1, n_2] + 2\sigma^2 \delta[n_1 - n_2] \end{aligned}$$

Marking Scheme:

- **3 Marks:** Correct simplification of $r_z[n_1, n_2]$ by expansion of all terms.
- **2 Marks:** Correct use of symmetry ($r_{xy} = r_{yx}$) and substitution.
- **1 Mark:** Final correct expression.

Note: Using $r_x[k]$ instead of $r_x[n_1, n_2]$ is not correct here as $x[n]$ is not specified as Wide Sense Stationary (WSS). Similarly, $y[n]$ is also not necessarily WSS.

Part b) [2 Marks]

Given: $x[n]$ and $y[n]$ are uncorrelated random processes.

- Because the processes are **uncorrelated and zero-mean**, the cross-correlation terms are zero:

$$r_{xy}[n_1, n_2] = E[x[n_1]]E[y[n_2]] = 0 \text{ and } r_{yx}[n_1, n_2] = 0$$

- **Final Expression:** The autocorrelation of $z[n]$ is the sum of the individual autocorrelations:

$$r_z[n_1, n_2] = r_x[n_1, n_2] + r_y[n_1, n_2]$$

Marking Scheme:

- **1 Mark:** Stating $r_{xy} = 0$ for uncorrelated zero-mean processes.
- **1 Mark:** Final correct expression.

Q2 [12 Marks]

Given: $x[n] = \phi + \cos(\pi n + \phi)$, where $\phi \sim U[0, 2\pi]$.

Part a) [4.5 Marks]

Three realizations based on sample values of ϕ , for example:

1. $\phi = 0$: $x[n] = \cos(\pi n) = (-1)^n$.
2. $\phi = \pi/2$: $x[n] = \pi/2 + \cos(\pi n + \pi/2) = \pi/2 - \sin(\pi n) = \pi/2$. ($\sin(\pi n) = 0$).
3. $\phi = \pi$: $x[n] = \pi + \cos(\pi n + \pi) = \pi - (-1)^n$. ($\cos(\pi n + \pi) = -\cos(\pi n)$)

Marking Scheme:

- **1.5 Marks** for each unique realization sketch, equations are not necessary.

Note:

- Not using x and y ticks/labels has attracted penalty.
- Drawing continuous time signals is incorrect as the focus is on discrete-time signals.

Part b) [5.5 Marks]

Mean :

$$\mu_x[n] = E[\phi] + E[\cos(\pi n + \phi)] = \int_0^{2\pi} \phi \frac{1}{2\pi} d\phi + 0 = \pi$$

Auto-correlation :

$$r_x[n, n+k] = E[\phi^2] + E[\phi \cos(\pi(n+k) + \phi)] + E[\phi \cos(\pi n + \phi)] + E[\cos(\pi n + \phi) \cos(\pi(n+k) + \phi)]$$

- **Term 1:** $E[\phi^2] = \frac{1}{2\pi} \int_0^{2\pi} \phi^2 d\phi = \frac{4\pi^2}{3}$
- **Terms 2 & 3:** Let α be a constant (πn or $\pi(n+k)$).

$$\begin{aligned} E[\phi \cos(\alpha + \phi)] &= \frac{1}{2\pi} \int_0^{2\pi} \phi \cos(\alpha + \phi) d\phi \\ &= \frac{1}{2\pi} [\phi \sin(\alpha + \phi) + \cos(\alpha + \phi)]_0^{2\pi} \\ &= \frac{1}{2\pi} [(2\pi \sin(\alpha + 2\pi) + \cos(\alpha + 2\pi)) - (0 + \cos(\alpha))] \\ &= \frac{1}{2\pi} [2\pi \sin(\alpha) + \cos(\alpha) - \cos(\alpha)] = \sin(\alpha) \end{aligned}$$

Since $\alpha = \pi n$ or $\pi(n+k)$ and $n, k \in \mathbb{Z}$, $\sin(\alpha) = 0$. Both cross terms vanish.

- **Term 4:**

$$\begin{aligned}
E[\cos(\pi n + \phi) \cos(\pi(n + k) + \phi)] &= E\left[\frac{1}{2} \cos(\pi n + \phi - \pi n - \pi k - \phi) + \frac{1}{2} \cos(2\pi n + \pi k + 2\phi)\right] \\
&= \frac{1}{2} E[\cos(-\pi k)] + \frac{1}{2} E[\cos(2\pi n + \pi k + 2\phi)] \\
&= \frac{1}{2} (-1)^k + 0 = \frac{1}{2} (-1)^k
\end{aligned}$$

Final Result: $r_x[n, n + k] = \frac{4\pi^2}{3} + \frac{1}{2}(-1)^k$

Marking Scheme:

- **1 Mark** for mean
- **1 Mark** for term 1 in auto-correlation function
- **2 Marks** for term 2&3
- **1.5 Marks** for term 4

Part c) [2 Marks]

A random process is defined as Wide Sense Stationary (WSS) if it satisfies the following three conditions:

1. The mean of the process is a constant ($\mu_x[n] = \mu$).
2. The autocorrelation depends only on the lag k ($r_x[n, n + k] = r_x[k]$).
3. The variance of the process is finite ($Var(x[n]) < \infty$ or $r_x[0] < \infty$).

Conclusion: Yes, the process is WSS because $\mu_x = \pi$, r_x depends only on k , and the variance is finite.

Marking Scheme:

- **1 Mark** for correctly stating all three conditions for WSS.
- **1 Mark** for the correct conclusion based on calculations in Part b).

Question 3

Part (a)

$$\begin{aligned}H(e^{j\omega}) &= \sum h[n]e^{-j\omega n} = 1 - \frac{1}{4}e^{-j\omega} \\|H(e^{j\omega})|^2 &= \left(1 - \frac{1}{4}e^{-j\omega}\right) \left(1 - \frac{1}{4}e^{j\omega}\right) = \frac{17}{16} - \frac{1}{2}\cos(\omega) \\S_y(e^{j\omega}) &= |H(e^{j\omega})|^2 S_v(e^{j\omega}) = \frac{17}{16} - \frac{1}{2}\cos(\omega), \quad (S_v = 1)\end{aligned}$$

Plot the curve

Rubric (Part a):

- Correct expression for $H(e^{j\omega})$ [1 mark]
- Correct computation of $|H(e^{j\omega})|^2$ [2 marks]
- Correct expression for $S_y(e^{j\omega})$ [1 mark]
- Correct plot of $S_y(e^{j\omega})$ [2 marks]

Part (b)

Method 1: Using IDTFT

$$\begin{aligned}S_y(e^{j\omega}) &= \frac{17}{16} - \frac{1}{4}e^{j\omega} - \frac{1}{4}e^{-j\omega} \\r_y[k] &= \frac{17}{16}\delta[k] - \frac{1}{4}\delta[k+1] - \frac{1}{4}\delta[k-1]\end{aligned}$$

Method 2: Using Direct Convolution

$$\begin{aligned}r_y[k] &= h[k] * h[-k] \\&= \delta[k] - \frac{1}{4}\delta[k-1] - \frac{1}{4}\delta[k+1] + \frac{1}{16}\delta[k] \\&= \frac{17}{16}\delta[k] - \frac{1}{4}\delta[k-1] - \frac{1}{4}\delta[k+1]\end{aligned}$$

From $r_y[0] = \frac{17}{16}$ and $r_y[\pm 1] = -\frac{1}{4}$, the matrix R_y is:

$$R_y = \begin{bmatrix} \frac{17}{16} & -\frac{1}{4} & 0 & 0 \\ -\frac{1}{4} & \frac{17}{16} & -\frac{1}{4} & 0 \\ 0 & -\frac{1}{4} & \frac{17}{16} & -\frac{1}{4} \\ 0 & 0 & -\frac{1}{4} & \frac{17}{16} \end{bmatrix}$$

Rubric (Part b):

- Correct $r_y[k]$ using **either** IDTFT **or** convolution [2.5 marks]
- Correct formation of R_y [1.5 marks]

Question 4

Part (a)

[Total: 6 Marks]

$$\begin{aligned} E[a] &= 0, & E[a^2] &= \frac{1}{3} \\ x[n] &= \begin{cases} a & n > 0 \\ 0 & n = 0 \\ -a & n < 0 \end{cases} \\ m_x[n] &= E[a(u[n] - u[-n])] = 0 \end{aligned}$$

Autocorrelation function:

$$\begin{aligned} R_{xx}[n_1, n_2] &= E[x[n_1]x[n_2]] \\ &= \begin{cases} \frac{1}{3}, & n_1 n_2 > 0 \\ -\frac{1}{3}, & n_1 n_2 < 0 \\ 0, & n_1 = 0 \text{ or } n_2 = 0 \end{cases} \end{aligned}$$

Rubric (Part a):

- Correct mean and variance of A [1 mark]
- Correct simplification of $x[n]$ [1 mark]
- Correct mean $m_x[n]$ [1 marks]
- Correct and simplified autocorrelation [3 marks]

Part (b)

[Total: 4 Marks]

Ensemble mean $m_x[n] = 0$

$$\text{Time average} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N a_0(u[n] - u[-n])$$

Sum = 0 (negative samples cancel positive symmetrically)

Time Average = Ensemble Mean $\implies$ Ergodic in mean

Rubric (Part b):

- Correct time average expression [1 mark]
- Correct evaluation of sum [2 marks]
- Correct justification [1 mark]

Q5

A WSS random process $x[n]$ has auto-correlation

$$r_x[k] = \alpha^{|k|} + \beta \delta[k].$$

Model $x[n]$ as an AR(2) process:

$$x[n] + a_1 x[n-1] + a_2 x[n-2] = v[n],$$

where $v[n]$ is white noise.

[2]

Yule–Walker equations for AR(2):

$$\begin{bmatrix} r_x[0] & r_x[1] \\ r_x[1] & r_x[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = - \begin{bmatrix} r_x[1] \\ r_x[2] \end{bmatrix}.$$

With $r_x[0] = 1 + \beta$, $r_x[1] = \alpha$, $r_x[2] = \alpha^2$:

$$\begin{bmatrix} 1 + \beta & \alpha \\ \alpha & 1 + \beta \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = - \begin{bmatrix} \alpha \\ \alpha^2 \end{bmatrix}.$$

[3]

Solving gives:

$$a_1 = -\frac{\alpha((1 + \beta) - \alpha^2)}{(1 + \beta)^2 - \alpha^2}, \quad a_2 = -\frac{\alpha^2 \beta}{(1 + \beta)^2 - \alpha^2}.$$

[3]

And the driving-noise variance relation is

$$\sigma_v^2 = r_x[0] + a_1 r_x[1] + a_2 r_x[2].$$

[2]